

Manuscript staging prose (§2–§5) — not ChatGPT Scholar VECTOR output

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Author: COMPASS (Cursor agent) — paragraphs to **copy from**, not read cover-to-cover
Ink map: 10_Manuscript_ink_map.md
Dakota (decision points): 11_Dakota_feedback_decision_points.md
Working outline (you edit): Manuscript_working_outline_v1.docx
Charles locks: ../3-Master_Plan/02_Charles_decisions_locked.md
Status: Staging prose for Summer–Fall 2026 submission track. Numbers from manifests at export time. **Copy FROM this file INTO Manuscript_working_outline_v1.docx** per ink map (#10). Apply Dakota digest fixes before **manuscript §4** (tenure) — especially staging §4 content, which maps to **manuscript §7** (Predictions), not manuscript §4.

§2 Theory and framing

2.1 Advancement under constrained distinction

Organizations advance a fraction of members through selective tournaments — promotion boards, draft slots, tenure cases — where evaluators compare individuals to **local peer reference groups**. When peer pools become elite, two forces coexist: stronger peers can raise development and signaling (**benefit**, B), and dense clusters of similarly strong, substitutable peers compress marginal returns to distinction (**constraint**, D). We summarize the net local environment as $L_{\text{net}} = B - D$ and study how advancement responds to leave-one-out (LOO) peer-quality proxies across three settings.

This framing sits between sociological accounts of **selective ecology and assortative matching** (Menger), which explain who enters elite pools, and **science-of-success** minimal-model discipline (Barabási, Wang, Gates), which asks for stylized facts, parsimonious mechanisms, and model-guided measurements — not curve-fitting alone.

2.2 Stylized fact (Rung 1)

We document a repeated **inverted-U**: advancement rates rise with LOO peer quality through mid-tier environments, then fall in the most elite tier. The pattern is established in **U.S. Army officer promotion** (Setting 1) and **NCAA men’s basketball draft selection** (Setting 2), with **preliminary** evidence in **R1 computer-science tenure** (Setting 3). We use associational language throughout; causal identification is not claimed.

2.3 Minimal mechanism (Rung 2 — basketball generative POC)

Beyond talent-only selection, a score that rewards own ability and penalizes LOO **viable-peer congestion** can produce non-monotone advancement versus team quality in simulation, while ability-only selection does not. Formally, selection uses $S_i = A_i - \lambda L_{C,\text{LOO}}$ with soft assignment into teams and top- K advancement. This operationalizes the **constraint leg D** inside the same $L_{\text{net}} = B - D$ ontology — not a separate model.

Mandatory limitation: The generative proof-of-concept is read on **whole-roster pool mean**; the empirical inverted-U is defined on **LOO pool quality** (poolq_loo). We do not claim bin-for-bin generative replication on the LOO axis in v1.

2.4 Model-guided measurements (Rung 2.5 — PD12 Priority 3)

Following Alex Gates (Paper Directions 12), the theory proposes **new quantities**, not only fits to legacy curves. We distinguish:

Quantity	Construct	Basketball column	Interpretation
Team quality	Mean peer performance	poolq_loo	LOO mean teammate ability
Viable-peer congestion	Density above prospect threshold	crowding_smooth	$C_{i,t}$ — constraint leg diagnostic

Elite environments raise both average talent and substitutability; the downturn should steepen where congestion is high, especially for **near-threshold** individuals, while top latent talent still clears selection.

§3 Methods — minimal model and measurements (basketball anchor)

3.1 Empirical Setting 2 (basketball)

Player-season panel with LOO pool quality (poolq_loo), draft outcome, and within-season performance controls. Figure 2: inverted-U on ventile bins of LOO pool quality (PPM z within-season). **Artifact:** scout_manuscript_v1/inverted_u_ventiles_ppm_zwithinseason_2026-06-24.png .

3.2 Generative POC (sports/538D_development.ipynb CELL 10 / 539 preset)

Frozen preset: Beta(2,2) ability, uniform team targets, crowding_smooth congestion, $\lambda = 0.55$, top- K selection. **Contrast panels:** ability-only (monotone) vs congestion-in-score (peak-and-decline on pool mean). **Artifacts:** generative_ability_only_pool_mean.png , generative_congestion_539_pool_mean.png , generative_congestion_539_loo_quality.png (limitation readout).

3.3 Axis table

See scout_manuscript_v1/axis_table_generative_readouts.md — maps model quantities to empirical columns per setting.

3.4 Settings 1 and 3 (empirical legs)

- **Army:** LOO senior-rater pool minus mean; CIF bar panels + cause-specific Cox (Fine–Gray deferred).
- **Tenure:** LOO department peer quality (`poolq_loo_mean`); stage 9 binned plot; inference sample **796 persons / 52 departments** (HIGH+MEDIUM OpenAlex match, `faculty_panel_inference_v1.csv`). Preliminary; Layer B Cox pre-submission.

§4 Predictions

4.1 Prediction #1 — Near-threshold heterogeneity (primary)

Statement: Holding own performance in the upper tail of the within-season distribution, draft probability should fall more sharply at the highest bins of LOO pool quality than for below-median performers — consistent with stronger distinction compression (D) where elite peer pools are both high-quality and congested.

Evidence status: Exploratory readout exported (`heterogeneity_ventiles_top_tail.png`). Named primary prediction for v1; not a pre-registered confirmatory test.

4.2 Prediction #2 — Peak shift with global Λ (prose hook)

Statement: Where the number of scarce advancement slots per local pool rises (higher Λ), the peak of the inverted-U should shift — Army board-size narrative (CODA-owned).

Evidence status: Prose hook for v1; empirical figure TBD. Not demonstrated in basketball generative sim for this manuscript version.

§5 Discussion and limitations

5.1 Contribution

We provide (1) a cross-domain empirical regularity on LOO peer quality, (2) a minimal generative ingredient for congestion in selection, (3) model-guided measurements separating quality from congestion, and (4) named prediction-facing readouts — packaged in a Wang-style ladder. The **development vs competitive constraint** distinction is largely absent from prior talent sociology and is explicit here.

5.2 Literature placement

- **Menger:** Selective ecology and assortative matching motivate elite pools; we perform the tri-domain comparison he discusses but does not run.
- **Barabási/Wang:** Performance vs success, feedback, and minimal models motivate structure — our outcome is **bounded advancement probability**, not superstar tail scaling.
- **Alex Gates (PD10–12):** Operational score, quality/congestion split, near-threshold emphasis.

5.3 Multiplicative vs additive model form (explicit deferral)

Menger and Wang discuss **multiplicative** skill bundles and log-normal success tails. v1 keeps an **additive selection score** and $B - D$ decomposition. Multiplicative production functions motivate **future** full $B(Q) - D(Q)$ generative work and **manuscript §5–§6** inequality discussion — **not** a v1 rewrite. See `05_Model_Nesting_Note_v1.md` §7.

5.4 Limitations

- Observational designs; no causal claims.
- Generative POC axis $\neq$ empirical LOO axis (honest).
- Tenure Setting 3 preliminary; attrition may conflate lateral moves.
- Army pool-size audit pending pre-publication.
- Predictions #1/#2 are candidates, not fully validated cross-domain results.

Appendix pointers

Item	Path
D10 bundle	<code>datasets/mbb/exports_inverted_u_v0/scout_manuscript_v1/</code>
Nesting note	<code>5-Manuscript/05_Model_Nesting_Note_v1.md</code>
Claim table	<code>3-Master_Plan/07_Claim_language_guardrails.md</code>
Tenure inference manifest	<code>tenure_pipeline/faculty_panel_inference_v1_manifest.json</code>

End VECTOR manuscript draft v1 sections.